

1 Solution — GIAM 2.5

Problem 4

1.1 Problem

A **Sophie Germain prime** is a prime p such that $2p + 1$ is also prime — for example 11, since $23 = 2 \cdot 11 + 1$ is prime. Almost all Sophie Germain primes are congruent to $5 \pmod{6}$, but there are exceptions, so the statement “There are Sophie Germain primes that are not $5 \pmod{6}$ ” is true. **Verify this.**

1.2 Answer

The statement is an **existential** claim, so it is verified by exhibiting a witness. There are exactly two:

$$\boxed{p = 2 \quad \text{and} \quad p = 3}$$

$$p = 2 : 2 \text{ is prime, } 2 \cdot 2 + 1 = 5 \text{ is prime, } 2 \equiv 2 \not\equiv 5 \pmod{6}.$$

$$p = 3 : 3 \text{ is prime, } 2 \cdot 3 + 1 = 7 \text{ is prime, } 3 \equiv 3 \not\equiv 5 \pmod{6}.$$

Both are Sophie Germain primes and neither is $5 \pmod{6}$, so the statement is true. ■

1.3 These Are the Only Two

The claim is an $\exists$ -statement, so one witness settles it. In fact there are exactly two, and no more.

The witnesses

p = 2 2 is prime ✓ $2 \cdot 2 + 1 = 5$ is prime ✓ $2 \bmod 6 = 2$, not 5	p = 3 3 is prime ✓ $2 \cdot 3 + 1 = 7$ is prime ✓ $3 \bmod 6 = 3$, not 5
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And these are the only two: for a Sophie Germain prime $p > 3$, every other residue is impossible

p mod 6	what goes wrong
0	6 divides p, so p is not prime
1	$2p+1 \equiv 3 \pmod{6}$, so 3 divides $2p+1$ — and $2p+1 > 3$, hence composite
2	p is even, so p is not prime ($p > 2$)
3	3 divides p, so p is not prime ($p > 3$)
4	p is even, so p is not prime
5	nothing goes wrong — the only surviving case

Residues 0, 2, 3, 4 fail because p would be composite; residue 1 fails because $2p+1$ would be.

So every Sophie Germain prime except 2 and 3 is congruent to 5 (mod 6).

Searching to 20000 turns up 326 of them, and only 2 and 3 sit outside the class — matching the proof exactly.

Figure 1.1: Top: the two witnesses. For p equal 2: 2 is prime, 2 times 2 plus 1 equals 5 is prime, and $2 \bmod 6$ equals 2, not 5. For p equal 3: 3 is prime, 2 times 3 plus 1 equals 7 is prime, and $3 \bmod 6$ equals 3, not 5. Below, a case analysis over the residue of p modulo 6, for a Sophie Germain prime p greater than 3. Residue 0: six divides p so p is not prime. Residue 1: $2p$ plus 1 is congruent to 3 mod 6, so 3 divides $2p$ plus 1, and since $2p$ plus 1 exceeds 3 it is composite. Residue 2: p is even so not prime since p exceeds 2. Residue 3: three divides p so not prime since p exceeds 3. Residue 4: p is even so not prime. Residue 5, highlighted: nothing goes wrong, the only surviving case. Conclusion: every Sophie Germain prime except 2 and 3 is congruent to 5 mod 6. A search to twenty thousand finds 326 of them, and only 2 and 3 lie outside the class, matching the proof.

The problem says “almost all,” which invites the sharper question: *how many* exceptions are there? The answer is **exactly two**, and this can be proved outright.

Claim. If p is a Sophie Germain prime with $p > 3$, then $p \equiv 5 \pmod{6}$.

Proof. Every integer is congruent to 0, 1, 2, 3, 4 or 5 modulo 6. Suppose $p > 3$ is prime and $2p + 1$ is prime, and rule out the first five residues:

- $p \equiv 0$: then $6 \mid p$, so p is not prime.
- $p \equiv 2$ or $p \equiv 4$: then p is even, and $p > 2$, so p is not prime.
- $p \equiv 3$: then $3 \mid p$, and $p > 3$, so p is not prime.

- $p \equiv 1$: then $2p + 1 \equiv 2 \cdot 1 + 1 = 3 \pmod{6}$, so $3 \mid 2p + 1$. Since $p > 3$ we have $2p + 1 > 3$, so $2p + 1$ is a proper multiple of 3 — composite, contradicting that p is Sophie Germain.

Only $p \equiv 5$ survives. ■

A search below 20,000 finds 326 Sophie Germain primes, and their residues mod 6 are exactly $\{2, 3, 5\}$ — the two exceptions plus the generic class, matching the proof.

1.4 Remark — Verifying an Existential Means Producing a Witness

Logically this problem is the simplest kind of task in the section. The statement has the form

$$\exists p \left[\text{SG}(p) \wedge p \not\equiv 5 \pmod{6} \right],$$

and an existential is verified by **exhibiting one object and checking it**. No general argument is needed and none would be better: $p = 2$ alone finishes the job. Contrast this with its negation, $\forall p [\text{SG}(p) \Rightarrow p \equiv 5]$, which is a universal and would require a proof covering *every* prime — exactly the asymmetry between $\exists$ and $\forall$ that Problems 2 and 3 kept exposing. Here the universal happens to be *nearly* true, which is what makes the exceptions worth naming.

1.5 Remark — Why $p \equiv 1$ Is the Interesting Case

Four of the five ruled-out residues fail for a trivial reason: p itself would be composite. Those cases say nothing about Sophie Germain primes in particular — they are just the standard fact that any prime $p > 3$ is coprime to 6, hence $p \equiv 1$ or 5. The **only** case doing real work is $p \equiv 1$, where p is a perfectly good prime but $2p + 1$ is killed by divisibility by 3. So the content of the theorem is precisely: *among primes coprime to 6, the 1 mod 6 ones are disqualified by their partner*. The smallest instances make this vivid — 7, 13, 19, 31 are all prime and all $\equiv 1$, and their partners 15, 27, 39, 63 are all divisible by 3.

1.6 Remark — Why 2 and 3 Escape

The two exceptions are exactly the primes that **divide** 6, and that is not a coincidence. The whole argument works modulo $6 = 2 \cdot 3$, and the steps " p even $\Rightarrow$ not prime" and " $3 \mid p \Rightarrow$ not prime" each carry the caveat $p > 2$ and $p > 3$. Those caveats are precisely where 2 and 3 slip through: 2 *is* even and prime, 3 *is* divisible by 3 and prime. This is a recurring pattern in modular arguments — the primes dividing the modulus are always the special cases, which is why theorems in number theory so often begin “let $p > 3$ ” or “for all primes not dividing n .”

1.7 Remark — The Congruence Is a Necessary, Not Sufficient, Condition

It is worth being clear about the direction of the result. Knowing $p \equiv 5 \pmod{6}$ does **not** make p a Sophie Germain prime; the congruence only rules out the other residues. Plenty of primes are $5 \pmod{6}$ without qualifying — 17 is prime and $17 \equiv 5$, but $2 \cdot 17 + 1 = 35 = 5 \cdot 7$ is composite; likewise 47, with $95 = 5 \cdot 19$. So the theorem is a *filter*, not a test: it narrows the search to one residue class but says nothing about which members of that class succeed. Confusing a necessary condition with a sufficient one is the converse error of Problem 2.2.9, dressed in number-theoretic clothing.

1.8 Remark — An Open Problem Lurking Nearby

Although we settled the mod-6 question completely, the basic question about Sophie Germain primes is **unsolved**: nobody knows whether there are infinitely many. Heuristics (and extensive computation — they have been found with tens of thousands of digits) strongly suggest yes, and there is a conjectured asymptotic count, but no proof exists. So this problem sits in a pleasant position: a statement that *looks* like it might be hard turns out to need only two small witnesses, while the neighbouring statement “there are infinitely many Sophie Germain primes” remains open. The primes themselves are named for Marie-Sophie Germain, who used them in early-nineteenth-century work on Fermat’s Last Theorem; they are also used in cryptography, where $2p + 1$ being prime gives the multiplicative group modulo $2p + 1$ a convenient structure.